

ELLIOT, South Africa

WMO: 686510

Lat: 31.3364S

Lon: 27.8403E

Elev: 1463

StdP: 84.94

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.2	-1.2	-13.7	1.4	15.0	-11.5	1.7	14.0	11.1	14.7	10.0	15.5	0.5	300	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	30.4	15.6	28.8	15.4	27.2	15.0	19.2	24.3	18.5	23.6	17.8	22.9	4.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	15.2	21.5	16.9	14.4	20.7	16.2	13.8	20.0	61.9	24.2	59.1	23.7	56.6	23.1	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.5	6.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	18.8	18.8	17.5	14.6	12.6	9.6	9.5	11.1	13.6	14.9	15.9	17.7
	DBStd	4.79	3.04	2.91	3.07	3.37	3.59	3.37	3.75	4.05	4.14	4.07	3.75	3.22
	HDD10.0	184	0	0	0	6	14	48	55	36	14	8	4	0
	HDD18.3	1576	31	26	51	116	181	263	273	227	149	118	89	52
	CDD10.0	1839	272	247	234	144	95	36	40	69	121	161	182	240
	CDD18.3	190	45	39	27	4	4	0	0	2	7	12	16	33
	CDH23.3	1802	349	270	206	48	30	0	0	18	126	224	198	332
CDH26.7	426	98	69	34	4	7	0	0	0	21	49	44	99	
Wind	WSAvg	2.9	2.9	2.6	2.5	2.1	2.6	3.2	3.0	3.1	3.1	3.2	3.2	3.2
Precipitation	PrecAvg	707	118	100	93	53	18	16	16	23	34	62	85	95
	PrecMax	1082	242	164	178	97	59	61	66	97	111	210	203	169
	PrecMin	220	48	30	27	13	1	2	1	0	2	9	11	27
	PrecStd	174	45	39	37	21	14	17	18	23	27	42	48	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	31.1	29.6	27.3	27.2	20.3	21.2	25.9	29.0	30.6	30.7	32.1
		MCWB	16.7	16.8	15.9	15.1	15.7	9.1	8.1	10.6	12.2	13.6	14.4	15.1
	2%	DB	29.4	29.1	27.5	24.8	23.4	18.7	19.4	23.3	26.6	28.1	28.0	29.4
		MCWB	16.7	17.0	15.7	13.6	12.9	8.1	7.9	9.7	11.1	13.1	14.6	15.7
	5%	DB	27.1	26.9	25.7	23.0	21.8	17.5	18.0	21.3	24.4	26.0	25.8	27.1
		MCWB	16.9	16.7	15.6	13.1	11.4	7.5	7.4	9.0	10.7	12.8	14.0	15.9
	10%	DB	24.9	24.5	23.9	21.2	19.8	16.1	16.5	19.4	22.3	23.7	23.6	24.7
		MCWB	16.8	16.7	15.1	12.9	10.2	7.1	7.1	8.5	10.4	12.1	14.2	16.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	20.2	19.3	17.1	18.6	10.9	9.9	12.7	15.0	16.4	17.6	20.0
		MCDB	26.1	24.6	24.1	23.6	23.7	17.2	16.1	19.6	21.9	23.9	23.5	24.5
	2%	WB	19.1	19.1	18.3	16.0	15.1	9.7	9.1	11.4	13.6	15.4	16.6	18.6
		MCDB	24.6	23.9	23.5	21.1	22.7	15.7	16.1	19.3	21.2	23.1	22.3	23.8
	5%	WB	18.3	18.3	17.4	15.0	12.5	8.8	8.3	10.4	12.6	14.4	15.9	17.7
		MCDB	23.4	23.2	22.6	20.3	19.6	14.9	15.7	18.5	20.2	21.8	21.7	22.9
	10%	WB	17.6	17.6	16.5	14.3	11.2	8.1	7.7	9.3	11.7	13.6	15.2	16.8
		MCDB	22.6	22.4	21.4	19.3	18.2	14.2	15.1	17.2	18.7	20.7	21.3	22.0
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.1	10.8	11.4	12.6	12.1	13.0	13.6	14.0	12.9	11.3	11.1
		MCDBR	14.7	14.6	14.3	14.6	15.1	13.4	14.2	15.7	17.8	18.2	16.6	16.7
		MCWBR	4.9	4.8	5.0	5.9	5.8	6.1	5.9	6.3	5.8	5.9	5.0	4.9
	5% WB	MCDBR	11.2	11.2	11.2	11.0	13.2	12.2	11.9	14.0	14.2	14.3	12.3	11.6
		MCWBR	5.5	5.2	5.0	5.1	6.1	6.4	5.6	6.2	5.8	5.9	5.3	5.7
Clear-Sky Solar Irradiance	taub	0.336	0.338	0.323	0.292	0.272	0.256	0.257	0.306	0.341	0.349	0.326	0.330	
	taud	2.484	2.478	2.496	2.563	2.589	2.600	2.583	2.429	2.330	2.352	2.471	2.496	
	Ebn at Noon	1003	982	964	946	920	915	929	914	928	959	1007	1012	
	Edh at Noon	116	113	104	87	75	70	74	98	121	127	117	116	
All-Sky Solar Radiation	RadAvg	6.68	6.11	5.32	4.36	3.58	3.19	3.48	4.19	5.23	5.96	6.69	6.92	
	RadStd	0.45	0.49	0.36	0.24	0.18	0.19	0.19	0.27	0.31	0.44	0.40	0.41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page